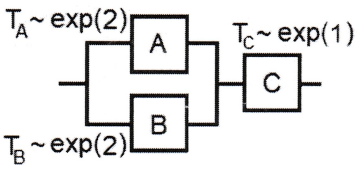
STATISTICS TICs - UAH - GROUP 2E1

Deliverable of problem **EN1** - October 21, 2022 - Duration of the test: 25 minutes

SURNAMES:_____ NAME:_____

Consider the system in the figure, in which the lifetimes of components A, B, and C are independent random variables of exponential type, with $T_A \sim \exp(\lambda_A = 2)$, $T_B \sim \exp(\lambda_B = 2)$ and $T_C \sim \exp(\lambda_C = 1)$, where the parameters of the exponential distributions represent the average number of failures per year. In this context:



a) *(0,5 pt)* Determine $F_T(x)$, that is, the cumulative distribution function (CDF) of the random variable $T \equiv$ *"lifetime of the complete system (in years)"*.

b) *(0.5 pt)* Determine $f_T(x)$ (the probability density function of the random variable T), and $E[T]$ (the expected value of T, that is, the average time (in years) that the entire system is expected to work before the first failure)

Remark: To obtain the maximum grade in part b), it is necessary to compute the integrals that appear using the tables.

(a) We consider the events: $(i = A, B, C)$

$(T_i > t) \equiv$ "The component i works (survives) up to instant t"

$(T > t) \equiv$ "The whole system works (survives) up to instant t"

The system works if, and only if, C works AND (at least one of the components A OR B work). Therefore the event $(T>t)$ can be expressed as follows:

$$(T>t) = [(T_A>t) \cup (T_B>t)] \cap T_C \qquad (1)$$

Having probabilities on both sides of (1) yields:

$$R_T(t) := P(T>t) = P([(T_A>t) \cup (T_B>t)] \cap (T_C>t)) \leftarrow T_A, T_B, \text{ and } T_C \text{ are independent random variables}$$

$$= P((T_A>t) \cup (T_B>t)) \cdot P(T_C>t)$$

RELIABILITY FUNCTION OF THE COMPLETE SYSTEM

$$P(T_A>t) + P(T_B>t) - P(T_A>t)\cdot P(T_B>t)$$

$$= \left(P(T_A>t) + P(T_B>t) - P(T_A>t)\cdot P(T_B>t)\right)\cdot P(T_C>t)$$

$$e^{-\lambda_A t} \quad e^{-\lambda_B t} \quad e^{-\lambda_A t} \quad e^{-\lambda_B t} \quad e^{-\lambda_C t}$$

$$= \left(e^{-2t} + e^{-2t} - e^{-4t}\right) e^{-t} = 2e^{-3t} - e^{-5t}, \quad t>0$$

Thus, $\boxed{F_T(t) = 1 - R_T(t) = 1 - 2e^{-3t} + e^{-5t}, \text{ for } t>0}$

(b) $\boxed{f_T(t) = \frac{\partial F_T(t)}{\partial t} = 6e^{-3t} - 5e^{-5t}, \text{ for } t>0}$ Hence,

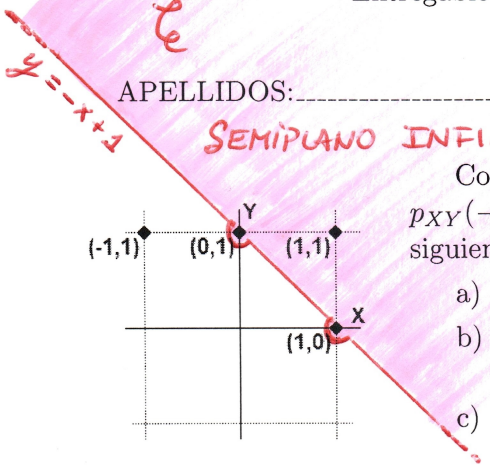
$$E[T] = \int_{S_T} t \cdot f_T(t)\, dt = \int_{t=0}^{t=\infty} t\cdot(6e^{-3t} - 5e^{-5t})\, dt = \int_{t=0}^{t=\infty} t\cdot 6e^{-3t}\, dt$$

$$- \int_{t=0}^{t=\infty} t\cdot 5e^{-5t}\, dt = 2\left(\frac{1}{3}\right) - \frac{1}{5} = \frac{7}{15} \simeq 0.46667 \text{ YEARS.}$$

ESTADÍSTICA TICs - UAH - GRUPO 2A1

Entregable **EN1** - 2 de noviembre de 2022 - Duración: 25 minutos

APELLIDOS:_____ NOMBRE:_____



SEMIPLANO INFINITO $X+Y \geq 1$ (para el apartado c)) $\mathcal{C}$

Considera el vector aleatorio discreto (X, Y) con función de probabilidad conjunta $p_{XY}(-1,1) = 3/8$, $p_{XY}(0,1) = p_{XY}(1,1) = 2/8$, $p_{XY}(1,0) = 1/8$. Contesta a las siguientes preguntas:

a) (*0.2 pt*) ¿Son independientes las variables X e Y ? Justifica tu respuesta.

b) (*0.5 pt*) Calcula el coeficiente de correlación de Pearson e interprétalo. ¿Son correladas o incorreladas las variables X e Y?

c) (*0.3 pt*) Calcula $P(X + Y \geq 1)$

(a) Los soportes marginales son $S_X = \{-1,0,1\}$ y $S_Y = \{0,1\}$, luego $S_X \times S_Y = \{(-1,0),(-1,1),(0,0),(0,1),(1,0),(1,1)\} \in \mathbb{R}^2$, y como $S_{XY} = \{(-1,1),(0,1),(1,1),(1,0)\}$ se observa que $S_{XY} \neq S_X \times S_Y$ y tenemos que **X e Y son DEPENDIENTES** (si hubiera salido $S_{XY} = S_X \times S_Y$ tendría que utilizar otro criterio)

(b) $\rho_{XY} = \dfrac{\sigma_{XY}}{\sigma_X \cdot \sigma_Y} = \dfrac{E[XY] - E[X]E[Y]}{\sqrt{V[X]} \cdot \sqrt{V[Y]}}$ con $-1 \leq \rho_{XY} \leq 1$, así que calculo todos los momentos que necesito.

Las funciones de probabilidad marginales son

En $S_X = \{-1,0,1\}$ tenemos $p_X(x) = \begin{cases} p_X(-1) = p_{XY}(-1,1) = \frac{3}{8} \\ p_X(0) = p_{XY}(0,1) = \frac{2}{8} \\ p_X(+1) = p_{XY}(1,1) + p_{XY}(1,0) = \frac{3}{8} \end{cases}$

En $S_Y = \{0,1\}$ tenemos $p_Y = \begin{cases} p_Y(0) = p_{XY}(1,0) = \frac{1}{8} \\ p_Y(1) = p_{XY}(-1,1) + p_{XY}(0,1) + p_{XY}(1,1) = \frac{7}{8} \end{cases}$

$$E[XY] = \sum_{(x,y) \in S_{XY}} x \cdot y \cdot p_{XY}(x,y) = (-1)(1)\cdot\frac{3}{8} + (0)(1)\cdot\frac{2}{8} + (1)(1)\cdot\frac{2}{8} + (1)(0)\cdot\frac{1}{8} = \frac{2}{8} - \frac{3}{8} = \frac{-1}{8}$$

$$E[X] = \sum_{x \in S_X} x \cdot p_X(x) = (-1)p_X(-1) + (0)p_X(0) + (+1)\cdot p_X(1) = \frac{-3}{8} + \frac{3}{8} = 0$$

$$E[Y] = \sum_{y \in S_Y} y \cdot p_Y(y) = (0)p_Y(0) + (1)\cdot p_Y(1) = \frac{7}{8}$$

$$E[X^2] = \sum_{x \in S_X} x^2 \cdot p_X(x) = (-1)^2 p_X(-1) + (0)^2 p_X(0) + (1)^2 p_X(1) = \frac{3}{8} + \frac{3}{8} = \frac{6}{8} = \frac{3}{4}$$

$$E[Y^2] = \sum_{y \in S_Y} y^2 p_Y(y) = (0)^2 p_Y(0) + (1)^2 p_Y(1) = \frac{7}{8}$$

Luego $V[X] = E[X^2] - (E[X])^2 = \frac{3}{4}$ y $V[Y] = E[Y^2] - (E[Y])^2 = \frac{7}{8} - \left(\frac{7}{8}\right)^2 = \frac{7}{64}$

FINALMENTE: $\rho_{XY} = \dfrac{(-1/8) - 0\cdot(7/8)}{\sqrt{3/4} \cdot \sqrt{7/64}} = \dfrac{-\sqrt{4}\cdot\sqrt{64}}{8\sqrt{3}\cdot\sqrt{7}} = \dfrac{-2\cdot 8}{8\sqrt{3}\sqrt{7}} = \dfrac{-2\sqrt{3}\sqrt{7}}{21} \approx -0.4364$

(c) $P(X+Y \geq 1) = \sum_{(x,y) \in \mathcal{C} \cap S_{XY}} p_{XY}(x,y) = p_{XY}(0,1) + p_{XY}(1,1) + p_{XY}(1,0) = \frac{2}{8} + \frac{2}{8} + \frac{1}{8} = \frac{5}{8}$

$\mathcal{C}$ es el semiplano infinito $X+Y \geq 1$, luego $\mathcal{C} \cap S_{XY} = \{(0,1),(1,1),(1,0)\}$, y la suma anterior se extiende a estos tres puntos de $\mathbb{R}^2$.

ESTADÍSTICA TICs - UAH - GRUPO 2B1

Entrega de problema **EN1** - 15 de noviembre de 2022 - Duración 25 minutos

APELLIDOS:_____ NOMBRE:_____

Considera el vector aleatorio (X, Y) con función de densidad conjunta $f_{XY}(x, y) = x + y$ en $S_{XY} = \{(x, y) \in \mathbb{R}^2 : (0, 1) \times (0, 1)\}$.

a) (0,2 pt) ¿Dirías que X e Y son independientes?

b) (0,4 pt) Calcula $P(Y < x \,|\, X = x)$ sabiendo que $P(Y \in \mathcal{C} \,|\, X = x) = \int_{\mathcal{C} \cap S_{Y|X=x}} f_{Y|X}(y|x)dy$

c) (0,4 pt) Utiliza el resultado anterior y el TPT para calcular $P(Y - X < 0)$

Ayuda: La expresión general del TPT en este caso es $P(Y \in \mathcal{C}) = \int_{S_X} P(Y \in \mathcal{C} \,|\, X = x) f_X(x)dx$ para todo conjunto $\mathcal{C} \in \mathbb{R}$

Solución

(a) Dado que $S_X \times S_Y = (0,1)\times(0,1) = S_{XY}$, este criterio no lleva a ninguna conclusión. X e Y son INDEPENDIENTES si, y solo si, $f_{XY}(x,y) = f_X(x) \cdot f_Y(y)$, luego calculamos las marginales y comprobamos:

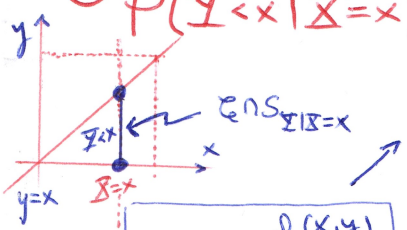
$$f_X(x) = \int_{S_{Y|X=x}} f_{XY}(x,y)\,dy = \int_{y=0}^{y=1} x+y\,dy = \int_{y=0}^{y=1} x\,dy + \int_{y=0}^{y=1} y\,dy = x\left[y\right]_{y=0}^{y=1} + \left[\frac{y^2}{2}\right]_{y=0}^{y=1} = x + \frac{1}{2},\ x \in (0,1)$$

$$f_Y(y) = \int_{S_{X|Y=y}} f_{XY}(x,y)\,dx = \int_{x=0}^{x=1} x+y\,dx = \int_{x=0}^{x=1} x\,dx + \int_{x=0}^{x=1} y\,dx = \left[\frac{x^2}{2}\right]_{x=0}^{x=1} + y\left[x\right]_{x=0}^{x=1} = y + \frac{1}{2},\ y \in (0,1)$$

Como $(x+\frac{1}{2})(y+\frac{1}{2}) = xy + \frac{1}{2}x + \frac{1}{2}y + \frac{1}{4} \neq x+y \Rightarrow$ X e Y: SON DEPENDIENTES

(b) $Y \in \mathcal{C}$

$$P(Y < x \,|\, X = x) = \int_{\mathcal{C} \cap S_{Y|X=x}} f_{Y|X}(y|x)\,dy = \int_{y=0}^{y=x} \frac{f_{XY}(x,y)}{f_X(x)}\,dy = \int_{y=0}^{y=x} \frac{x+y}{x+1/2}\,dy =$$



$$= \frac{1}{x+\frac{1}{2}} \int_{y=0}^{y=x} x+y\,dy = \frac{1}{x+\frac{1}{2}} \left(x\left[x\right]_{y=0}^{y=x} + \left[\frac{y^2}{2}\right]_{y=0}^{y=x} \right) = \frac{3x^2}{2\left(x+\frac{1}{2}\right)}$$

NOTA: $f_{Y|X}(y|x) = \dfrac{f_{XY}(x,y)}{f_X(x)}$

(c) DEL APARTADO ANTERIOR

$$P(Y - X < 0) = P(Y < X) = \int_{S_X} P(Y < x \,|\, X = x) f_X(x)\,dx =$$

$$= \int_{x=0}^{x=1} \frac{3x^2}{2\left(x+\frac{1}{2}\right)} \cdot \left(x+\frac{1}{2}\right) dx = \frac{3}{2} \int_{x=0}^{x=1} x^2\,dx = \frac{3}{2}\left[\frac{x^3}{3}\right]_{x=0}^{x=1} =$$

$$= \frac{3}{2}\left[\frac{1}{3}\right] = \frac{1}{2}$$